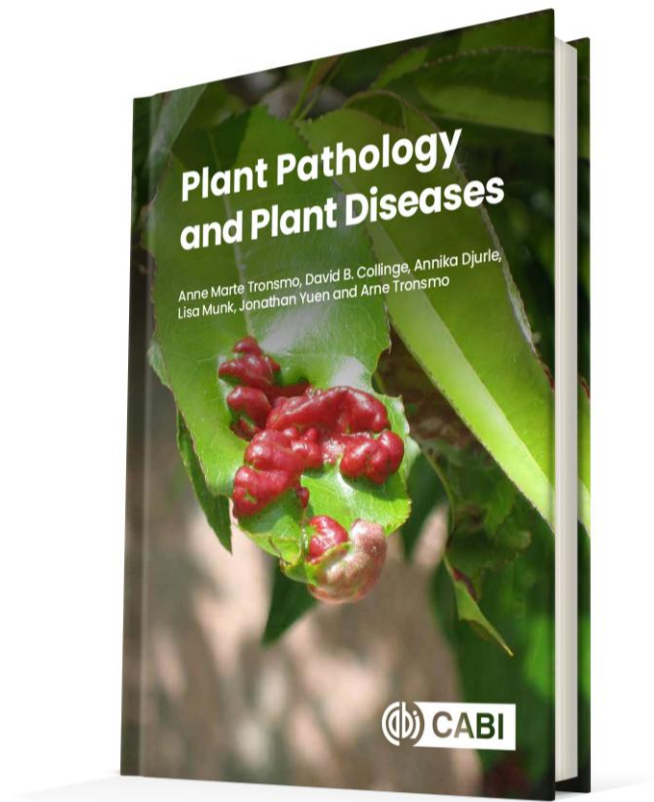




Plant Pathology and Plant Diseases

© Anne Marte Tronsmo, David B. Collinge, Annika Djurle,
Lisa Munk, Jonathan Yuen and Arne Tronsmo 2020

Fig. 10.1 Two different strategies for colonizing a host – biotrophy and necrotrophy. (A) Bean rust (*Uromyces viciae-fabae*) on *Vicia faba*. (B) Stemphylium blight (*Stemphylium botryosum*) on *Vicia faba*. (C) Barley powdery mildew (*Blumeria graminis*) and (D) scald (*Rhynchosporium graminicola*) both on *Hordeum vulgare*. (A) and (C). The rust and powdery mildew fungi have penetrated the host and are sporulating freely with no visible host reaction. (B) and (D) These necrotrophic pathogens have killed the host tissue and colonized the dead tissue. (A, B, © P. Taylor; C, D, © D.B.Collinge.)

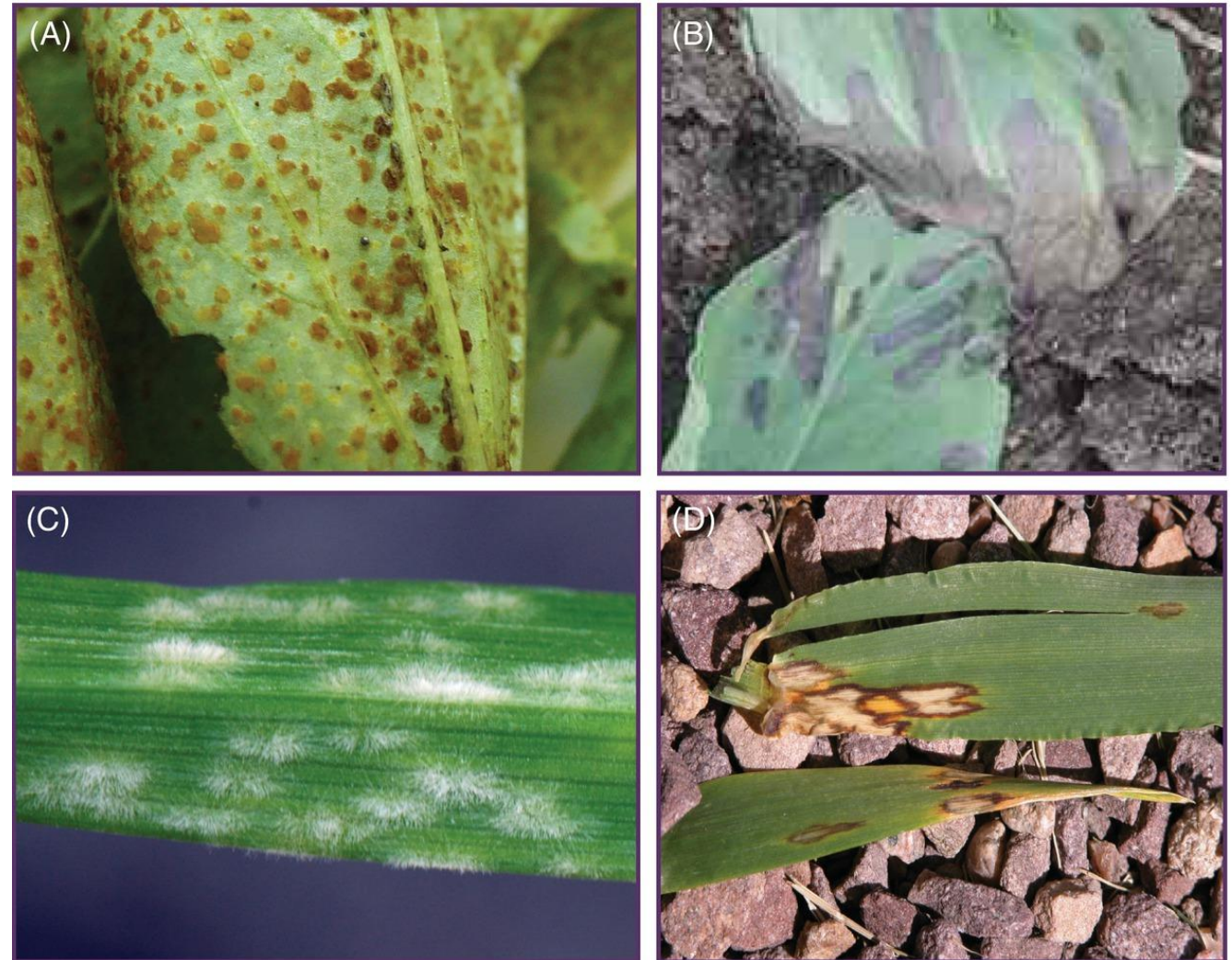
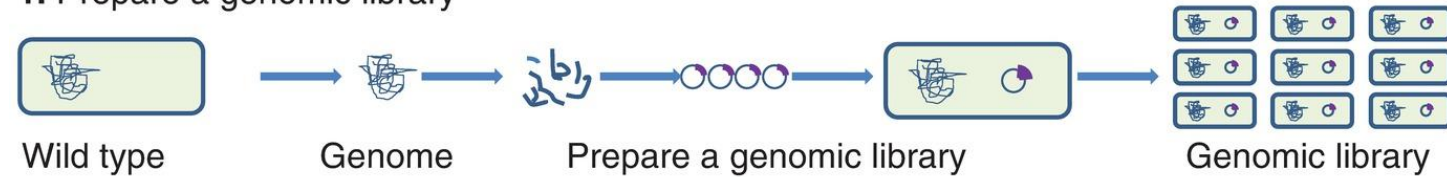


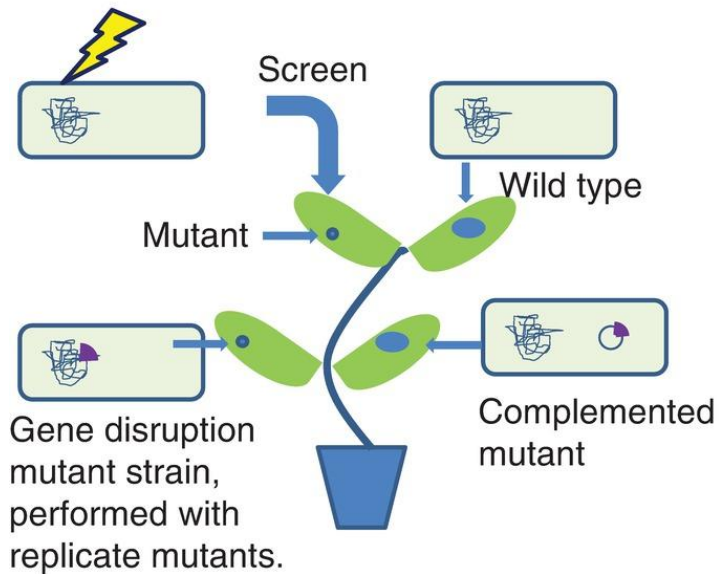
Fig. 10.2 Scheme for the identification and characterization of pathogenicity mutants.

(© D.B.Collinge.)

1. Prepare a genomic library



2. Mutagenize, screen for mutants and complementation test using a pathogenicity test



3. Identification of the mutated gene by subcloning, saturation mutagenesis and gene sequencing.



4. Gene disruption by recombination or site-directed mutagenesis (including CRISPR-Cas)

5. Physiological tests *in vitro* to confirm identity using wild type, mutant



Fig. 10.3 Mechanisms of pathogenicity related to main lifestyle strategies of pathogens.
(© D.B.Collinge.)

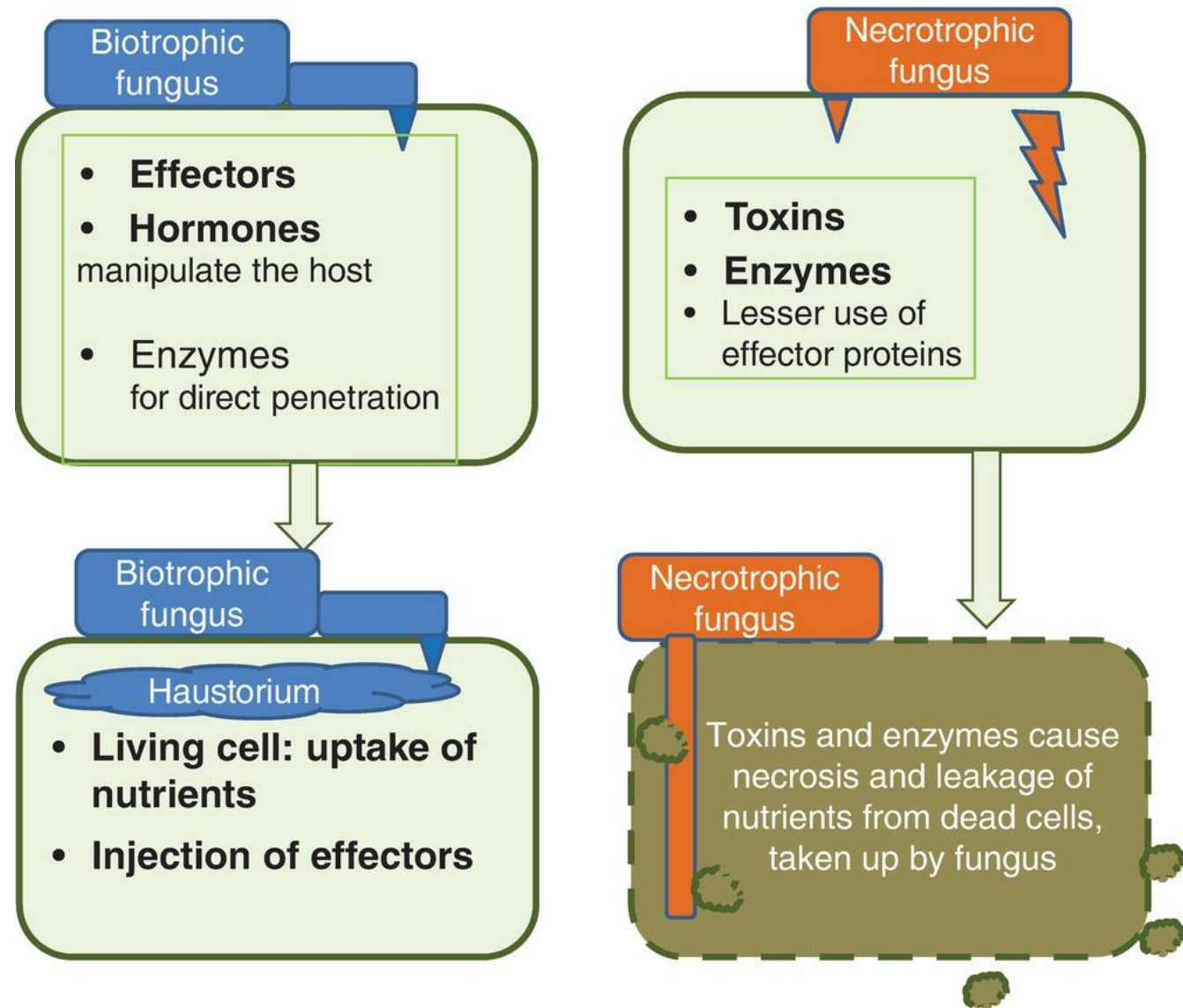


Fig. 10.4 Structure of a plant cell wall. (A) Structure of a cell wall illustrating the different components: cellulose fibrils, pectin, proteins. AGP, Arabinogalactan-proteins. (B) A pectin structure. (© J.U. Fangel.)

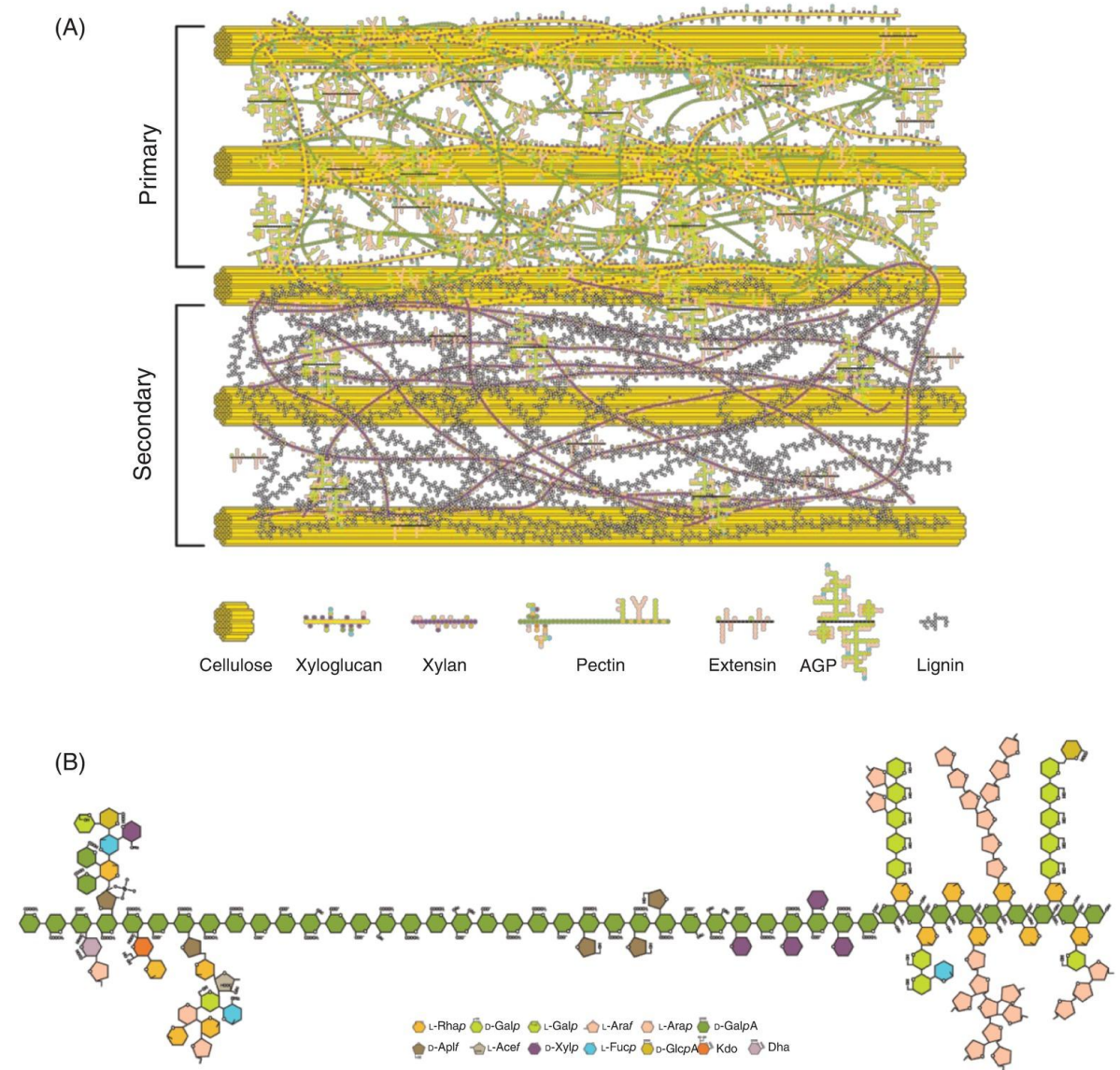


Fig. 10.5 Chemical structures of some important phytotoxins. (© D.B.Collinge.)

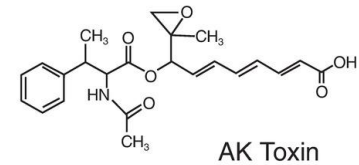
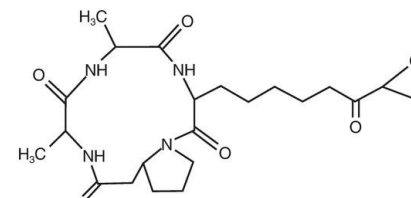
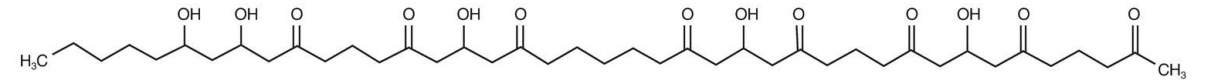
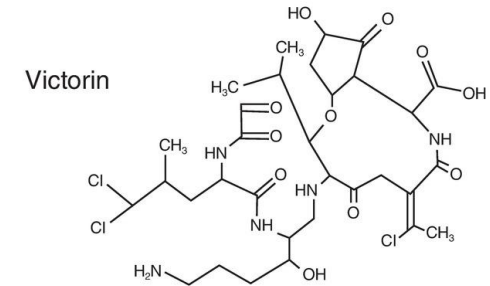
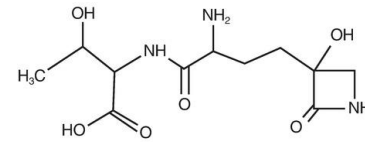
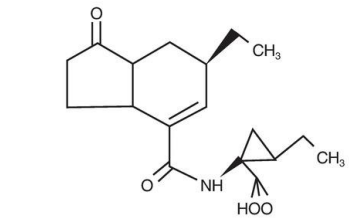
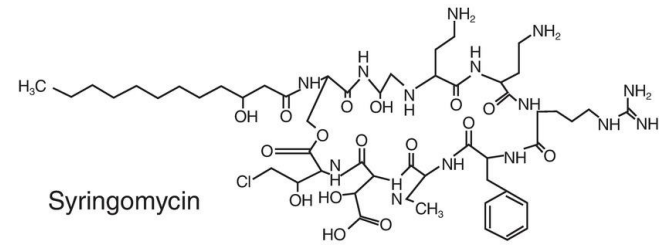
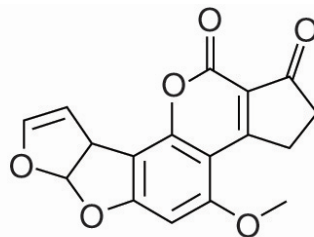
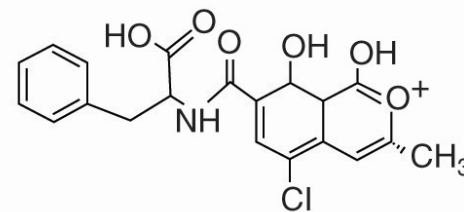


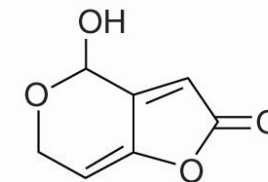
Fig. 10.6 Chemical structures of some important mycotoxins. (© D.B.Collinge.)



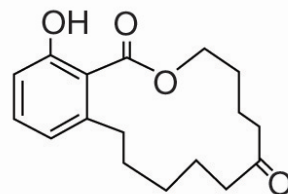
Aflatoxin B1



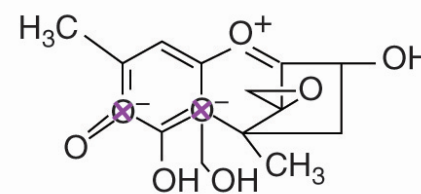
Ochratoxin A



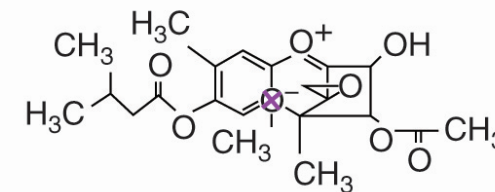
Patulin



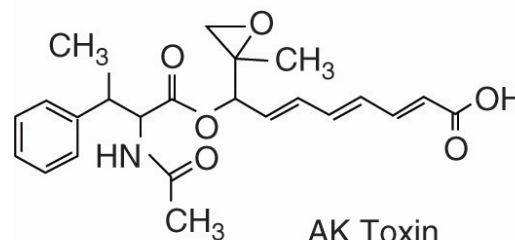
Zearalenone



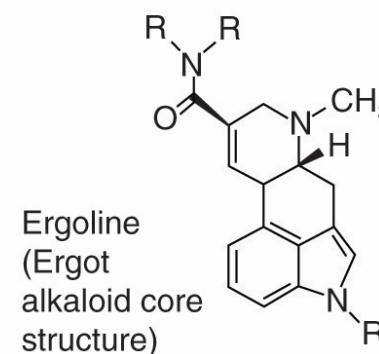
Deoxynivalenol
(Type B Tricothecene)



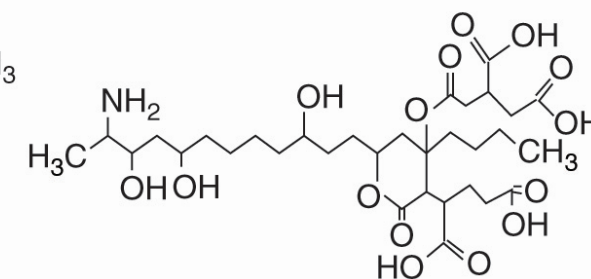
Tricothecene T2 (Type A)



AK Toxin



Ergoline
(Ergot
alkaloid core
structure)



Fumonisin B1

Fig. 10.7 A-D Examples of hormones in

pathogenicity. (A) *Taphrina deformans* causes peach leaf curl by manipulating auxin levels.

(B) *Taphrina betulina* causes witch's broom on birch.

(C) *Moniliophthora perniciosa* on cocoa.

(D) *Ustilago maydis* causes maize smut by manipulating cytokinins and abscisic acid levels.

(E) The aecia of *Gymnosporangium sabinae* on pear contain enhanced levels of cytokinins.

(F) Green islands are a characteristic symptom reducing senescence associated with rust and mildew infections. Result of a cytokinin effect.

(G) The gall wasp *Cynips quercusfolii* causes galls on oaks (H)

Gall on Jamaican sorrel (*Hibiscus sabdariffa*)

caused by *A. tumefaciens*. (A, B, D,: ©

D.B.Collinge; C: © P. Taylor.)

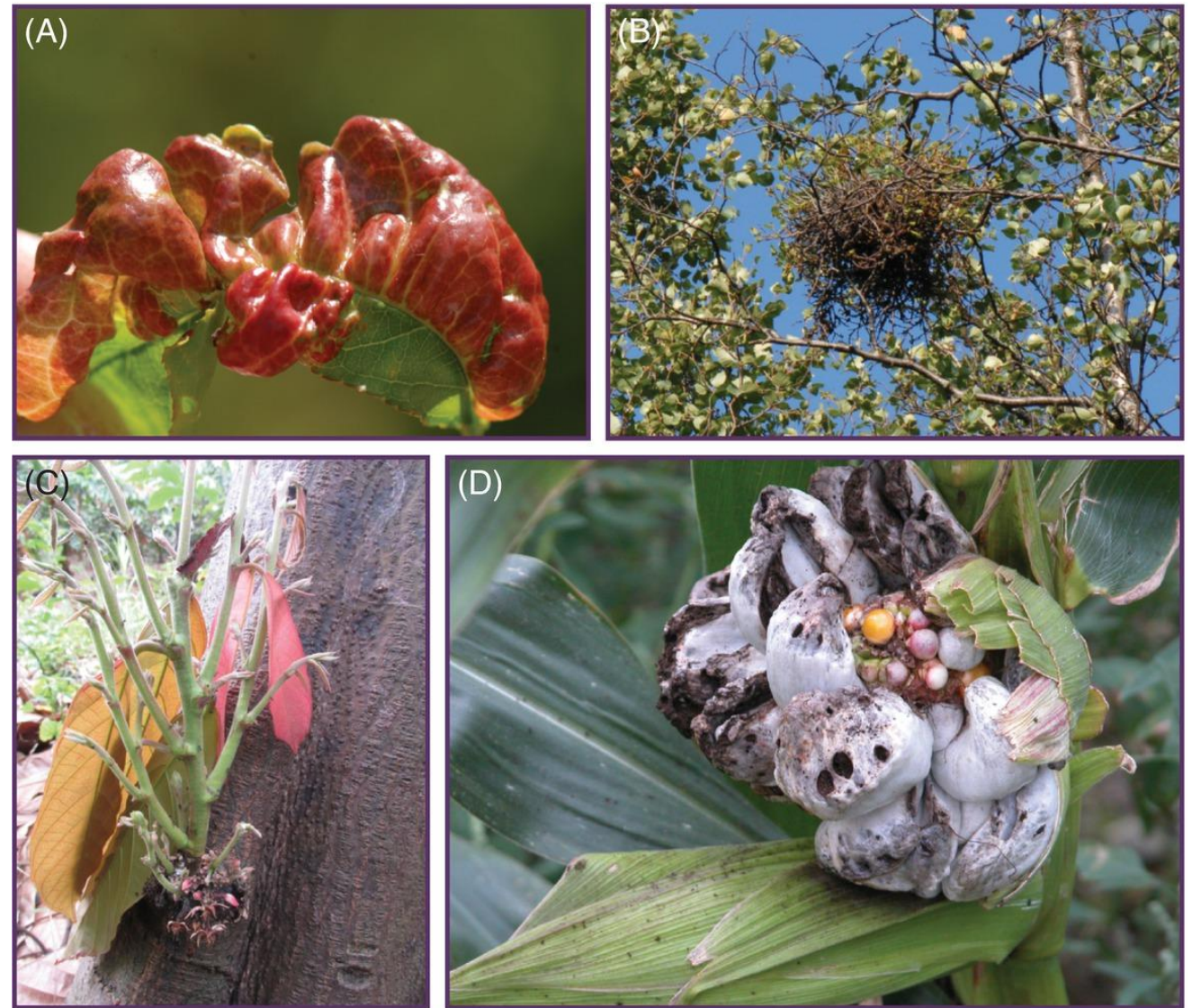


Fig. 10.7 E-H Examples of hormones in pathogenicity. (E) The aecia of *Gymnosporangium sabinae* on pear contain enhanced levels of cytokinins. (F) Green islands are a characteristic symptom reducing senescence associated with rust and mildew infections. Result of a cytokinin effect. (G) The gall wasp *Cynips quercusfolii* causes galls on oaks (H) Gall on Jamaican sorrel (*Hibiscus sabdariffa*) caused by *A. tumefaciens*. (E,F,G: © D.B.Collinge; H: © P. Taylor.)

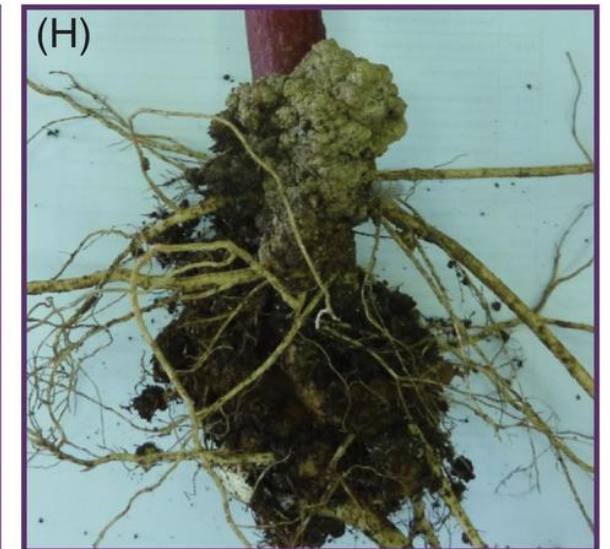


Fig. 10.8 Infection by *Agrobacterium*

***tumefaciens*.** *Agrobacterium tumefaciens* causes crown gall in strains carrying the Ti plasmid which transfers genes for the production of the hormones indole acetic acid (IAA, an auxin) and cytokinins to stimulate tumourgenesis. (© D.B.Collinge.)

